

Ota Naevus Treatment and Crusting

Crusts typically develop within the day following laser irradiation and may persist for up to 10 days. To minimize the risk of scarring, a minimum interval of 2 months between treatment sessions on the same area is recommended. The number of laser passes required to achieve complete clinical clearance of an Ota naevus depends on the density and depth of the dermal melanocytosis. Generally, more treatment sessions are needed for lesions in the intra-orbital region compared to other facial areas.

The introduction of picosecond lasers, a newer generation of quality-switched (QS) lasers, has not significantly altered the overall prognosis for Ota naevus treatment. However, these lasers may offer a shorter recovery period and potentially reduce the incidence of hypopigmentation. Further prospective studies are needed to confirm these observations.

Recommendation

The **1064 nm QS laser** is the gold standard for the treatment of both Ota and Ito naevi.

Acquired Dermal Melanocytosis

Acquired dermal melanocytosis presents as blue-grey or brown macules or patches that appear during puberty or adulthood. The exact pathophysiology is unclear, but reactivation of pre-existing dermal melanocytes is the most widely accepted theory. This condition is more commonly reported in individuals of Asian descent and occurs more frequently in women.

Potential triggering factors include ultraviolet (UV) exposure, trauma, inflammatory dermatoses, and pregnancy, although spontaneous onset is also common. Lesions can occur anywhere on the body, but the face is the most frequently affected site. When facial lesions are bilateral and symmetrical, they are classified as Acquired Bilateral Nevus of Ota-like Macules (ABNOM).

This condition is often mistaken for melasma, making accurate diagnosis critical. Unlike melasma, acquired dermal melanocytosis does not respond to skin-lightening agents or chemical peels. However, it shows excellent response to treatment with **1064 nm QS lasers**.

Laser and Light Treatment

Evidence on the treatment of acquired dermal melanocytosis is limited, but both clinical literature and experience indicate that lesions respond well to QS lasers, particularly at **1064 nm** or **755 nm**, often requiring fewer sessions than congenital forms.

Initial reports using the **755 nm alexandrite QS laser** noted a high rate (50%) of secondary dyschromia. In contrast, more recent studies using the **1064 nm QS laser** in a total of 44 patients reported only two cases of transient post-inflammatory hyperpigmentation (PIH) and no cases of hypopigmentation.

Recommendation

Acquired dermal melanocytosis should be recognized by clinicians as a distinct entity from melasma. The **1064 nm QS laser** is highly effective and the treatment of choice.

Congenital Nevomelanocytic Nevi (CNN)

Congenital nevomelanocytic nevi (CNN) are benign proliferations of neural crest-derived melanocytes, present at birth due to in utero mutations. They are classified based on predicted adult size (PAS):

- Small: <1.5 cm PAS
- Medium: 1.5–20 cm PAS
- Large: 20–40 cm PAS
- Giant: >40 cm PAS

Additional features assessed include the number of satellite nevi, location, color variation, surface texture, presence of dermal or subcutaneous nodules, and hypertrichosis.

Larger CNN size and a higher number of satellite nevi are strongly associated with increased risk of developing CNN syndrome (formerly neurocutaneous melanosis) and malignant transformation to melanoma.

Management decisions are based on multiple factors, including risk of malignancy, functional impact, aesthetic concerns, psychosocial effects, surgical complexity, and risks related to sedation and anesthesia.

Recommended Management

- Regular surveillance for melanoma
- Baseline and follow-up photographic documentation
- Periodic neurodevelopmental assessments
- Comprehensive patient and family education